# Introduction to Statistics — Quiz #2 (60 minutes)

**October 20, 2026 (Tuesday)**

Section: _____ Cadet Number: _____ Name: _____ Score:

- All solutions must include a detailed step-by-step explanation.
- If an answer has more than four decimal places, round to the **fourth decimal place**.
- **Reference table** is provided on the last page of the exam.

1. A multiple-choice test has $10$ questions, and each question has $5$ choices. A student picks every answer at random, so the probability of getting each question right is $1/5$. Let $X$ be the number of questions the student gets right out of the $n = 10$ questions. Answer the following questions. [30 points]

(1) Find the probability mass function $f(x)$ of $X$.

Solution: Each question is answered correctly with probability $1/5 = 0.2$, so $X \sim B(10,\, 0.2)$ and its probability mass function is

$$f(x) = \begin{cases} \binom{10}{x}(0.2)^x(0.8)^{10-x}, & x = 0, 1, \ldots, 10, \\ 0, & \text{otherwise.} \end{cases}$$

(2) Find the probability that the student gets at least one question right, $P(X \geq 1)$.

Solution: Since the probability mass function sums to 1, $P(X \geq 1) = 1 - P(X = 0)$. As $X \sim B(10,\, 0.2)$,

$$P(X \geq 1) = 1 - P(X = 0) = 1 - \binom{10}{0}(0.2)^0(0.8)^{10} = 1 - 0.1074 = \boxed{0.8926}.$$

(3) Find the expectation and variance of the random variable $X$.

Solution: For $X \sim B(n, p)$, we have $E(X) = np$ and $\mathrm{Var}(X) = np(1-p)$. Since $X \sim B(10,\, 0.2)$,

$$E(X) = 10 \times 0.2 = \boxed{2}, \qquad \mathrm{Var}(X) = 10 \times 0.2 \times 0.8 = \boxed{1.6}.$$

2. A robot shoots at a target. Each shot scores $1$, $2$, $3$, or $4$ points, and each of these four scores has probability $1/4$. The scores of different shots are independent. Answer the following questions. [30 points]

(1) Let $X_1$ and $X_2$ be the scores of the first and second shots. Find the probability mass function $f(\bar{x})$ of the average score $\bar{X} = (X_1 + X_2)/2$. (Fill in the table below.)

| $\bar{x}$ | | | | | | | |
|---|---|---|---|---|---|---|---|
| $f(\bar{x})$ | | | | | | | |

Solution: $f(1) = \frac{1}{16}$, $f(1.5) = \frac{1}{8}$, $f(2) = \frac{3}{16}$, $f(2.5) = \frac{1}{4}$, $f(3) = \frac{3}{16}$, $f(3.5) = \frac{1}{8}$, $f(4) = \frac{1}{16}$.

(2) Find $E(\bar{X})$ and $\mathrm{Var}(\bar{X})$.

Solution: Since $X_1$ and $X_2$ have the same distribution, $E(\bar{X}) = \{E(X_1) + E(X_2)\}/2 = E(X_1)$, and

$$E(X_1) = 1 \times \tfrac{1}{4} + 2 \times \tfrac{1}{4} + 3 \times \tfrac{1}{4} + 4 \times \tfrac{1}{4} = \boxed{2.5}.$$

Also, $E(X_1^2) = 1^2 \times \frac{1}{4} + 2^2 \times \frac{1}{4} + 3^2 \times \frac{1}{4} + 4^2 \times \frac{1}{4} = \frac{30}{4} = 7.5$, so $\mathrm{Var}(X_1) = E(X_1^2) - \{E(X_1)\}^2 = 7.5 - 2.5^2 = 1.25$. Since $X_1$ and $X_2$ are independent and have the same distribution,

$$\mathrm{Var}(\bar{X}) = \frac{\mathrm{Var}(X_1) + \mathrm{Var}(X_2)}{4} = \frac{\mathrm{Var}(X_1)}{2} = \frac{1.25}{2} = \boxed{0.625}.$$

3. The lifetime of portable flashlights made at a factory has mean $\mu = 2{,}000$ hours and standard deviation $\sigma = 50$ hours. Let $X_1, X_2, \ldots, X_n$ be the lifetimes (in hours) of $n = 100$ flashlights randomly selected from this factory. Find the probability that their sample mean $\bar{X}$ is not greater than $1{,}990$ hours, $P(\bar{X} \le 1{,}990)$. [20 points]

   Solution: $X_1, X_2, \ldots, X_n$ is a random sample of size $n = 100$ from a population with mean $\mu = 2{,}000$ and standard deviation $\sigma = 50$.

   Since the sample size $n = 100$ is large enough, by the central limit theorem,

$$\bar{X} \;\dot{\sim}\; N\left(2{,}000,\ \frac{50^2}{100}\right) = N\left(2{,}000,\ 5^2\right).$$

   Therefore,

$$\begin{aligned} P(\bar{X} \le 1{,}990) &= P\left(\frac{\bar{X} - 2{,}000}{5} \le \frac{1{,}990 - 2{,}000}{5}\right) \\ &= P(Z \le -2) = P(Z \ge 2) = \boxed{0.0228}, \end{aligned}$$

   since $z_{0.0228} = 2$.

4. The time a bus takes to reach its last stop follows a normal distribution with standard deviation $\sigma = 1$ minute. A random sample of $n = 17$ buses is taken and their travel times are recorded. Find the probability that the sample variance $S^2$ is greater than 2, $P(S^2 > 2)$. [20 points]

   Solution: $S^2$ is the sample variance of the travel times of the $n = 17$ selected buses.

   Since the population is normal,

$$V = \frac{(n-1)S^2}{\sigma^2} \sim \chi^2(n-1), \quad \text{so} \quad V = \frac{(17-1)S^2}{1^2} = 16S^2 \sim \chi^2(16).$$

   Therefore,

$$P(S^2 > 2) = P(16S^2 > 16 \times 2) = P(V > 32) = \boxed{0.01},$$

   since $\chi^2_{0.01}(16) = 32.00$.

**Reference Table** For $Z \sim N(0,1)$ and $V \sim \chi^2(n)$, let $z_\alpha$ and $\chi^2_\alpha(n)$ be the values satisfying $P(Z > z_\alpha) = \alpha$ and $P(V > \chi^2_\alpha(n)) = \alpha$.

| | | | |
|---|---|---|---|
| $z_{0.4207} = 0.2$ | $z_{0.3446} = 0.4$ | $z_{0.3085} = 0.5$ | $z_{0.1587} = 1$ |
| $z_{0.0668} = 1.5$ | $z_{0.0228} = 2$ | $z_{0.0062} = 2.5$ | $z_{0.0013} = 3$ |
| $\chi^2_{0.99}(16) = 5.81$ | $\chi^2_{0.05}(16) = 26.30$ | $\chi^2_{0.025}(16) = 28.85$ | $\chi^2_{0.01}(16) = 32.00$ |
| $\chi^2_{0.99}(17) = 6.41$ | $\chi^2_{0.05}(17) = 27.59$ | $\chi^2_{0.025}(17) = 30.19$ | $\chi^2_{0.01}(17) = 33.41$ |